assign-day-4

July 6, 2023

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1 Assignment Problems: Day 4

The following problems require access to the file named 'Dataset_Day4.csv', provided with this assignment. About the data, * Grades from a Chemical Engineering course at McMaster University. * The recorded values are the average of sub-components: e.g the Tutorial variable is the average of all tutorials, the Final exam variable is the average of all questions in the final, written exam. * The Prefix column is the year in which the student first enrolls at the university and is a crude approximation of the student's age (maturity). * This particular course permitted students to work in groups for assignments, tutorials and the take-home exam. The groups were self-selected, and varied during the semester. * Of interest is whether the assignments, tutorials, midterms or take-home exam are a good predictor of the student's performance in the final exam. Also, does the Prefix variable show any promise as a prediction variable?

[2]:
```
data = pd.read_csv("Dataset_Day4.csv")
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Prefix      99 non-null     int64
 1   Assignment  99 non-null     float64
 2   Tutorial    99 non-null     float64
 3   Midterm     99 non-null     float64
 4   TakeHome    98 non-null     float64
 5   Final       96 non-null     float64
dtypes: float64(5), int64(1)
memory usage: 4.8 KB
```

2 question - 1

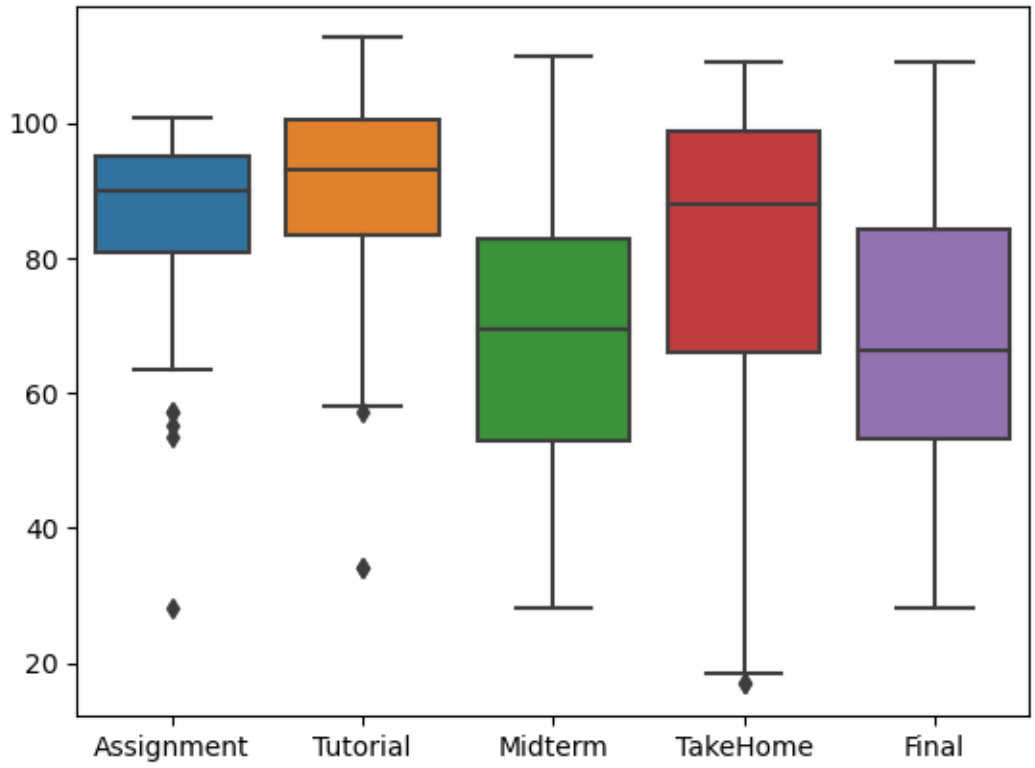
2.0.1 Find and treat all the missing values.

2.0.2 Rows or columns with missing values must not be dropped.

[3]:
```
sns.boxplot(data = data[['Assignment', 'Tutorial', 'Midterm', 'TakeHome',␣
 ↪'Final']])
data.skew(),data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Prefix      99 non-null     int64
 1   Assignment  99 non-null     float64
 2   Tutorial    99 non-null     float64
 3   Midterm     99 non-null     float64
 4   TakeHome    98 non-null     float64
 5   Final       96 non-null     float64
dtypes: float64(5), int64(1)
memory usage: 4.8 KB
```

[3]:
```
(Prefix       -1.746493
 Assignment   -1.676609
 Tutorial     -1.472330
 Midterm      -0.063458
 TakeHome     -1.175746
 Final         0.096656
 dtype: float64,
 None)
```

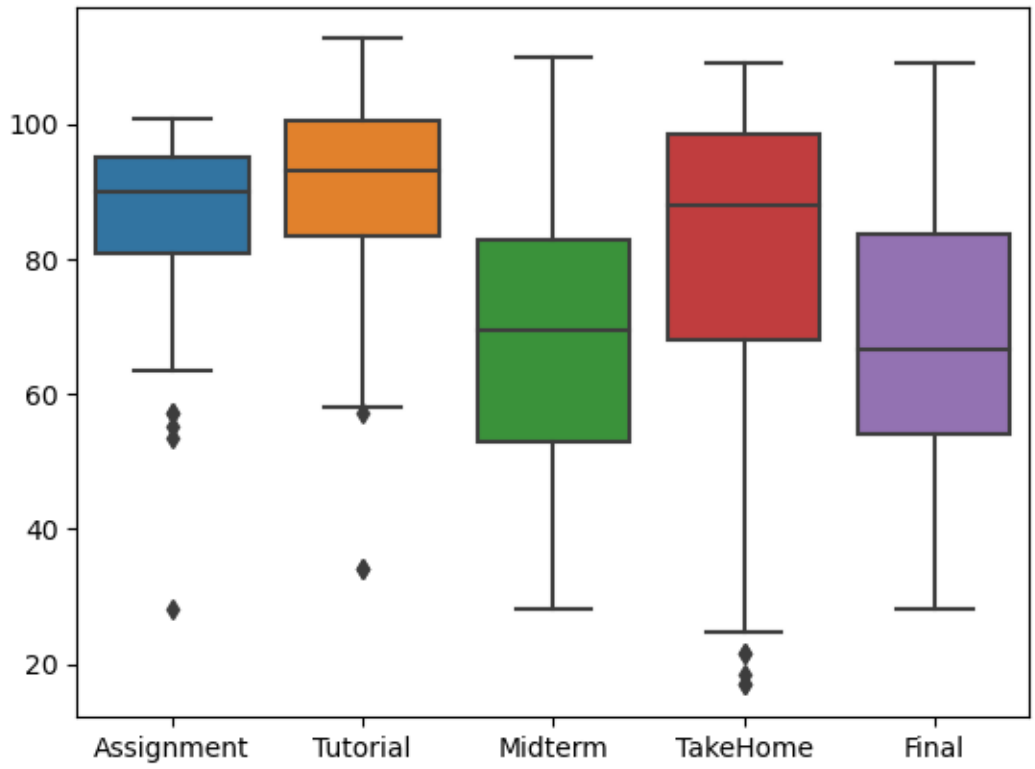


[4]:
```
data_nonull = data.copy()
# TakeHome is negetively skewed hence we will use median to replace the null␣
 ↪values
# Final is very slightly skewed ( almost zero ) hence we can use mean to␣
 ↪replace null values
data_nonull["TakeHome"].fillna(data_nonull["TakeHome"].median(),inplace=True)
data_nonull["Final"].fillna(data_nonull["Final"].mean(),inplace=True)
sns.boxplot(data = data_nonull[['Assignment', 'Tutorial', 'Midterm',␣
 ↪'TakeHome', 'Final']])
data.info(),data_nonull.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Prefix      99 non-null     int64
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 3   Midterm     99 non-null     float64
 4   TakeHome    98 non-null     float64
 5   Final       96 non-null     float64
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```
dtypes: float64(5), int64(1)
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<class 'pandas.core.frame.DataFrame'>
RangeIndex: 99 entries, 0 to 98
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
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 3   Midterm     99 non-null     float64
 4   TakeHome    99 non-null     float64
 5   Final       99 non-null     float64
dtypes: float64(5), int64(1)
memory usage: 4.8 KB
```

[4]: (None, None)

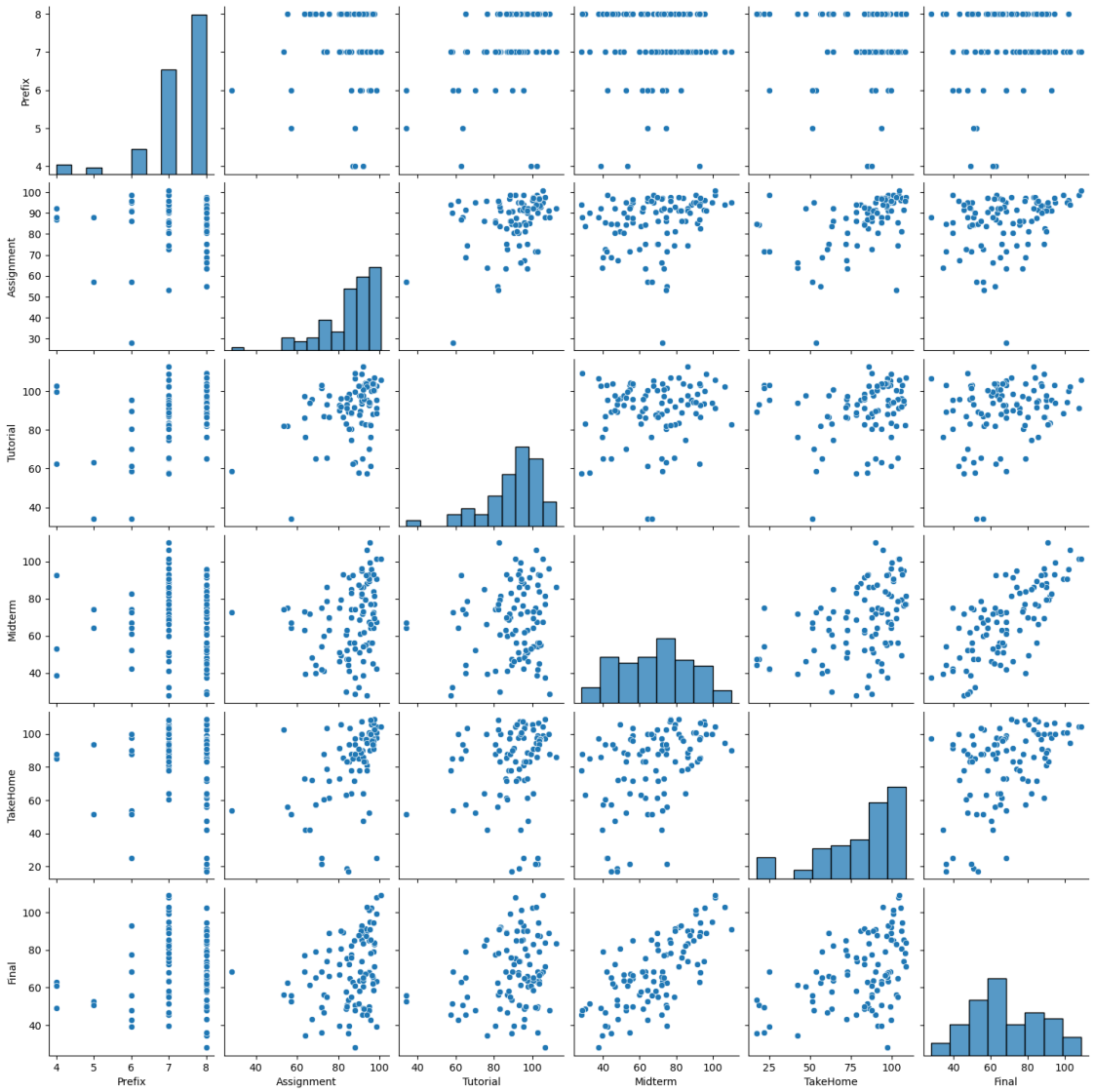


3 question - 2

3.0.1 Make an in-depth report on the variables using descriptive statistics and also portray the relationship between each variables using visualizations.

[5]:
```
sns.pairplot(data_nonull)
```

[5]:
```
<seaborn.axisgrid.PairGrid at 0x2a0dd038210>
```



[6]:
```
data_nonull.corr()
```

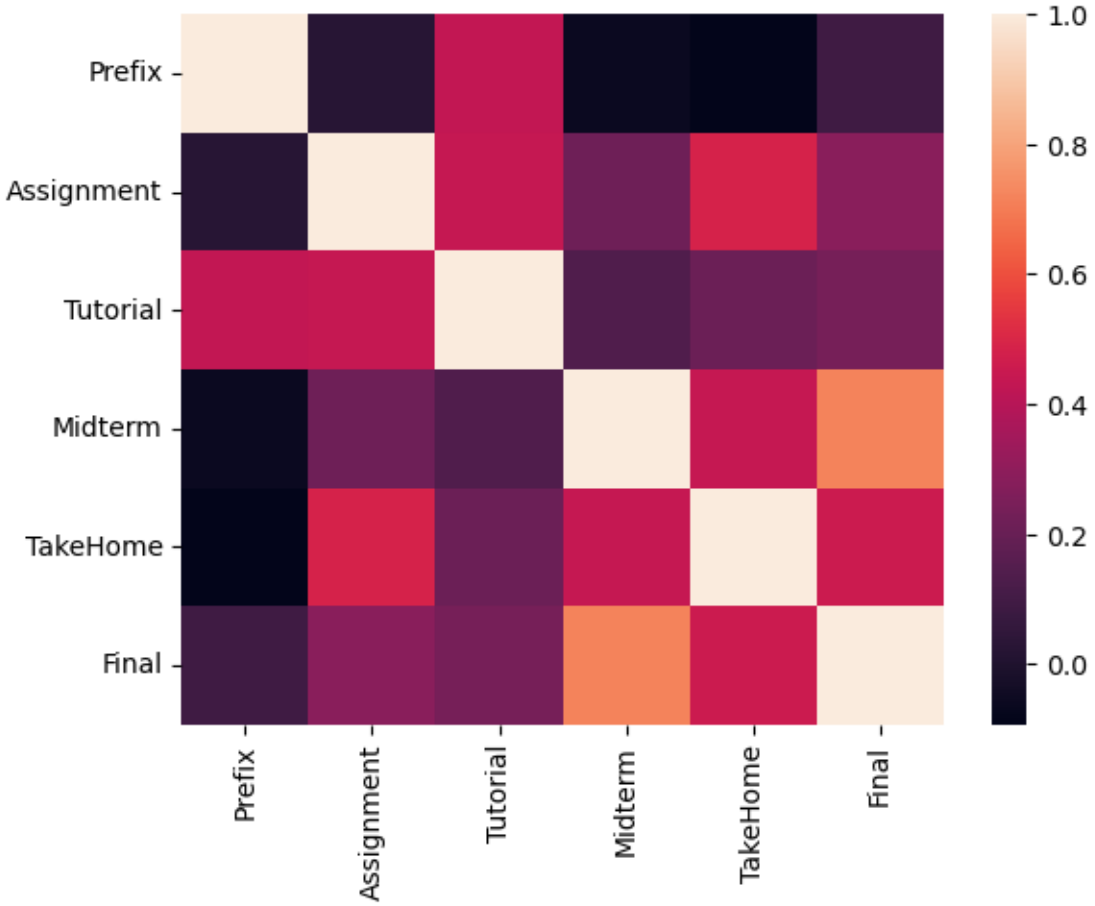
[6]:

	Prefix	Assignment	Tutorial	Midterm	TakeHome	Final
Prefix	1.000000	0.022476	0.431078	-0.062544	-0.092604	0.088832

```
Assignment   0.022476    1.000000  0.440115  0.215868  0.486430  0.284477
Tutorial     0.431078    0.440115  1.000000  0.135597  0.208791  0.239544
Midterm     -0.062544    0.215868  0.135597  1.000000  0.440537  0.717490
TakeHome    -0.092604    0.486430  0.208791  0.440537  1.000000  0.456543
Final        0.088832    0.284477  0.239544  0.717490  0.456543  1.000000
```

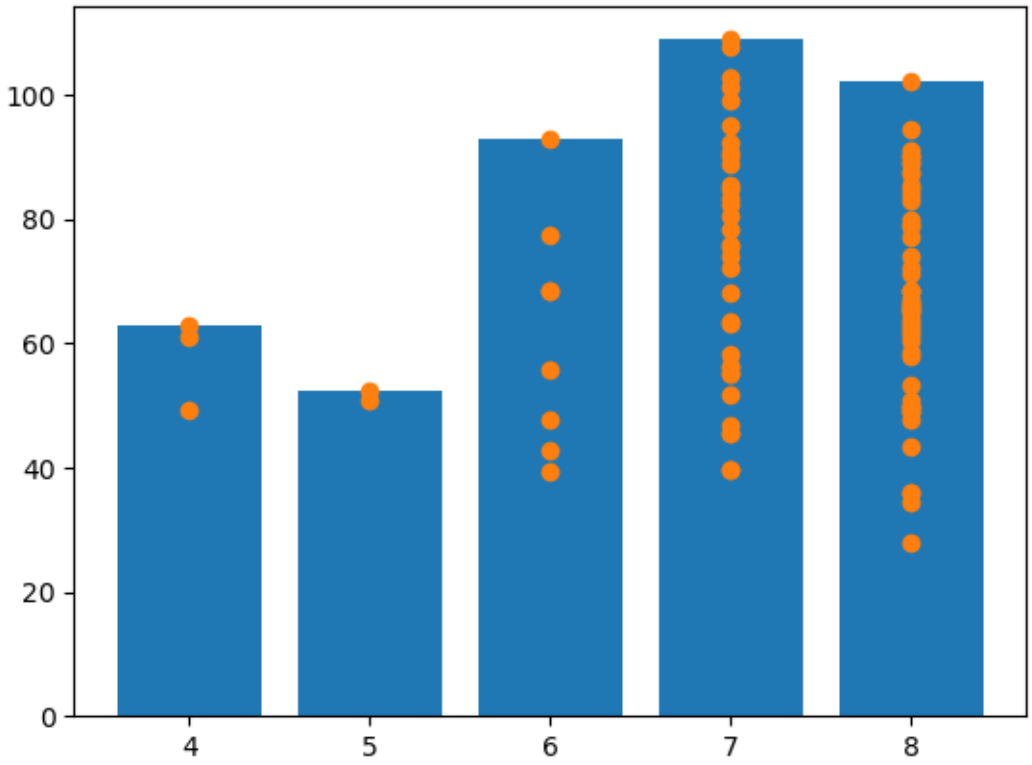
[7]:
```
sns.heatmap(data_nonull.corr())
```

[7]: <Axes: >



[8]:
```
plt.bar(data_nonull.Prefix, data_nonull.Final)
plt.scatter(data_nonull.Prefix, data_nonull.Final)
```

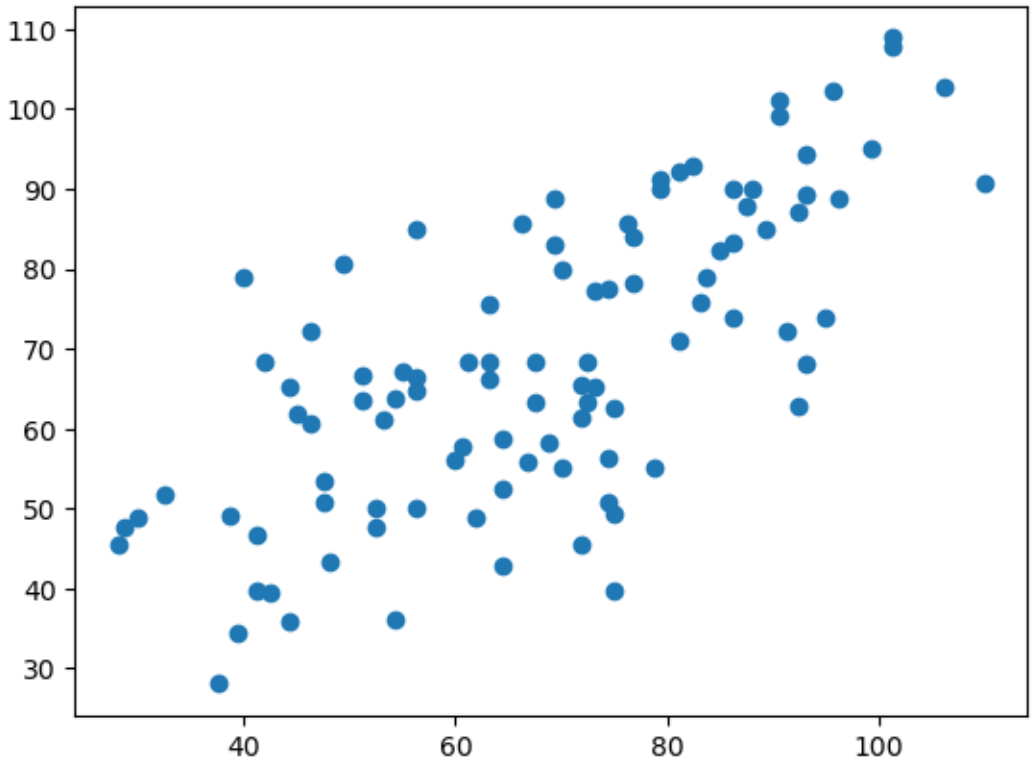
[8]: <matplotlib.collections.PathCollection at 0x2a0dee3e750>



- So we can observe from our prefix,final bar graph, scatter graph that on an average people with higher prefix have scored higher marks in Final

[9]:
```
plt.scatter(data_nonull.Midterm, data_nonull.Final)
```

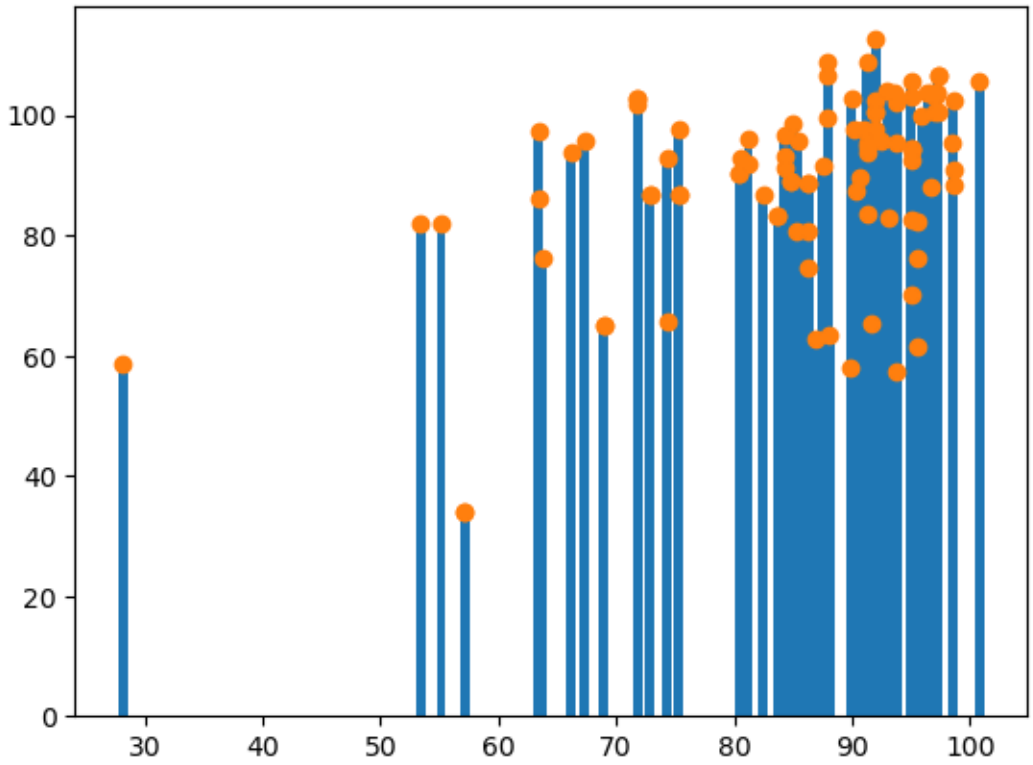
[9]:
```
<matplotlib.collections.PathCollection at 0x2a0df8b2790>
```



- In the above scatter graph, linear correlation hot , linear correlation table of Midterm and Final result seems almost linearly relatable
- We can observe this linear relation by joining a line from lower left part of the graph to the upper left part of the graph pssing through the plotted point

[10]:
```
plt.bar(data_nonull.Assignment, data_nonull.Tutorial)
plt.scatter(data_nonull.Assignment, data_nonull.Tutorial)
```

[10]:
```
<matplotlib.collections.PathCollection at 0x2a0df8f63d0>
```



- From the above graph between Assignment and Tutorial we can say that most of the people who did good in Tutorials did good in assignments too
- Most of the people did good in Assignment and Tutorial

3.0.2 Name - Nikhil Khatri

3.0.3 SIC - 21BCSA92